

INDEPENDENT PEER-REVIEW PACKET

EC-006

Planar Truncated-Normal Second-Moment Identity

Cold Review Packet 2 of 20

Dylan M. Roy Mathematical Research Program

Prepared for independent validation - 27 July 2026

This packet is self-contained for mathematical review. Drive access is supplemental, not required to understand the claim and proof.

READ FIRST

You are being asked to review one narrow terminal object, not the wider q0/P0.2/Theorem B research program. The declared terminal label is an internal research-state classification. Your task is to decide whether the exact statement and its proof/evidence are correct within the stated hypotheses and exclusions.

1. Pass A - Independent reconstruction. Read the exact statement, notation, and hypotheses. Attempt your own derivation or falsification before reading the embedded historical review record.
2. Pass B - Source audit. Compare your reconstruction with the complete controlling source text embedded in Part II. Check every dependency, sign, factor, quantifier, scope boundary, and computational claim.
3. Pass C - Hostile review. Run the object-specific focal checks and invent at least one additional mutation or counterexample attempt.
4. Issue one fail-closed verdict: APPROVE; APPROVE WITH CLARIFICATIONS; AMEND REQUIRED; REJECT; or CANNOT VERIFY.

Do not transfer approval to excluded downstream claims. Do not count access to this packet, the author, or another reviewer as independent mathematical evidence.

Part I - Exact Review Object

Packet	2 of 20
Object ID	EC-006
Object class	Exact Gaussian moment identity
Exact statement	For Q distributed as $N(-b, 2)$, $z_0(b) = E[Q^2 \mathbf{1}\{Q < 0\}] = (b^2 + 2) \Phi(b/\sqrt{2}) + \sqrt{2}b \phi(b/\sqrt{2})$; at $b=6/5$ the supported decimal is approximately 3.230978535287005.
Internal terminal label	CLOSED - EXACT PLANAR TRUNCATED-NORMAL SECOND-MOMENT IDENTITY
Terminal date	2026-07-21
Recommended expertise	Probability / Gaussian integrals / numerical analysis

1. Significance

This object was separated from larger theorem claims so that its correctness can be assessed without accepting unrelated Gaussian-field, persistence, adjacency, or asymptotic assertions. A positive review supports only the exact statement above and the dependency uses explicitly listed below.

2. Evidence class and prior review

Evidence	Closed-form integration, independent quadrature, Monte Carlo corroboration, and precision/scope firewall.
Prior review record	Independent formula derivation and numerical checks were performed across multiple lines.
Binding exclusions	The decimal is planar, not the exact side-24 torus coefficient; no finite-r or Palm theorem follows.

3. Dependency map and use restrictions

- Inputs: only definitions, hypotheses, and subordinate identities explicitly reproduced in the controlling source text.
- Output: only the exact statement listed in the review-object table.
- Permitted downstream use: as a cited lemma/interface under identical object definitions and hypotheses.
- Forbidden transfer: no approval of a parent theorem, numerical constant, external release, or canonical promotion unless separately reviewed.
- Object-specific exclusions: The decimal is planar, not the exact side-24 torus coefficient; no finite-r or Palm theorem follows.

4. Object-specific hostile checks

- Derive the truncated second moment without using the supplied formula.
- Check the sign of the linear phi term for mean -b.
- Reproduce the decimal independently and state the precision actually supported.
- Attempt at least one additional mutation that changes an orientation, sign, normalization, law, endpoint, or quantifier while leaving the text superficially similar.

5. Proof-obligation ledger

Obligation	Reviewer test	Initial state
O1 Object identity	Definitions, ordering, normalization, and law match the statement.	OPEN FOR REVIEW
O2 Hypotheses	Every hypothesis used is stated and no stronger premise is hidden.	OPEN FOR REVIEW
O3 Derivation	Each algebraic, analytic, topological, or probabilistic step is valid.	OPEN FOR REVIEW
O4 Quantifiers	Uniformity, existence, endpoint, and asymptotic quantifiers are correct.	OPEN FOR REVIEW
O5 Computation	Any executable or interval result is reproducible and non-vacuous.	OPEN FOR REVIEW
O6 Scope firewall	The conclusion is not silently enlarged beyond the exact object.	OPEN FOR REVIEW
O7 Contradictions	No known correction, counterexample, or retained caveat defeats the claim.	OPEN FOR REVIEW

Part II - Complete Controlling Source Text

The following is a paragraph-order-preserving text reproduction of the controlling Drive artifact. Typography is normalized for this packet; mathematical content is not intentionally rewritten. Use the manifest URL for native-file or byte-level checks.

GP-CLS-PROP-007-v1.0 — PROPOSED TERMINAL CLOSURE OF THE EXACT PLANAR $z_0(6/5)$ NORMALIZER FORMULA

Artifact ID: GP-CLS-PROP-007-v1.0

Author: OpenAI GPT-5.6 Thinking

Created: 2026-07-21

Class: CLS-PROP — additive terminal-closure proposal

Status: READY FOR HUMAN APPROVAL

Authority: none until Dylan M. Roy accepts or modifies the terminal statement

Canonical impact: NONE before approval

Queue item: EC-006

1. EXACT OBJECT TO BE CLOSED

Let Q be the planar coalesced curvature coordinate with

$$Q \sim N(-b, 2),$$

and define

$$z_0(b) = E[Q^2 \mathbf{1}\{Q < 0\}].$$

The exact planar truncated-normal identity is

$$\begin{aligned} z_0(b) &= (b^2 + 2) \Phi(b/\sqrt{2}) \\ &\quad + \sqrt{2} b \phi(b/\sqrt{2}), \end{aligned}$$

where Φ and ϕ are the standard normal distribution and density functions. At $b=6/5$,

$$\begin{aligned} z_0(6/5) &= (86/25) \Phi(3\sqrt{2}/5) \\ &\quad + (6\sqrt{2}/5) \phi(3\sqrt{2}/5) \\ &= 3.2309785352870049480962465690181631507648580105587... \end{aligned}$$

Proposed terminal label:

CLOSED — EXACT PLANAR TRUNCATED-NORMAL SECOND-MOMENT IDENTITY

2. FIRST-PRINCIPLES DERIVATION

Write $Q = \mu + \sigma Z$ with $\mu = -b$, $\sigma = \sqrt{2}$, $Z \sim N(0, 1)$, and $\alpha = (0 - \mu)/\sigma = b/\sqrt{2}$. For a Gaussian $X \sim N(\mu, \sigma^2)$, direct integration by parts gives

$$E[X^2 \mathbf{1}\{X \leq a\}]$$

$$= (\mu^2 + \sigma^2) \Phi(\alpha)$$

$$- \sigma(\mu + a) \phi(\alpha),$$

with $\alpha = (a - \mu)/\sigma$. Setting $a=0$, $\mu=-b$, $\sigma=\sqrt{2}$ gives

$$E[Q^2 \mathbf{1}\{Q < 0\}]$$

$$= (b^2 + 2) \Phi(b/\sqrt{2})$$

$$+ \sqrt{2} b \phi(b/\sqrt{2}).$$

No asymptotic argument, simulation, or unproved theorem is used in this identity.

3. EVIDENCE AND INDEPENDENT REVIEW

Primary/source-level statement:

- 05_RELIABILITY_AND_EVIDENCE.md §2 states the same exact planar formula and numerical value for $Q \sim N(-b, 2)$.

Independent CL re-execution:

- CL-DATA-026-v1.0, Drive ID 1bJmhmwgOYNQs7Ky737CT_fwJg7oSAhQvuKzDtVCFAMg, independently evaluates the closed form as 3.230978535 and reports a 2,000,000-draw Monte Carlo value 3.23094. It explicitly labels the result CONFIRMED-BY-REEXECUTION.

Independent CW route:

- CW-AUD-017-v1.0, Drive ID 1hFlr4t8cc2b7O92enQ_Hi1m7LYVFEUgI, verifies the normalizer reduction line by line and reports that the same decimal is obtained independently through the coalesced Palm-denominator battery. It describes two derivations yielding one constant.

Independent AO48 census:

- AO48-REG-018-v1.0, Drive ID 1gvZWNOWQDoU3Ec1lBvDW1HXcQSYBQRhs, lists the formula as kill-ready with an exact derivation, AO48 closed-form check, Monte Carlo band, and CW two-route match.

OpenAI scope audit and fresh derivation:

- This closure proposal re-derives the formula from the general Gaussian truncated second moment and independently evaluates the decimal to 50 digits.

4. LOAD-BEARING SCOPE LIMIT

This closure applies only to the planar coalesced law $Q \sim N(-b, 2)$.

It does NOT assert that the exact normalized-periodized side-24 torus coefficient is literally the same decimal. GP-DER-044-v1.0 — Exact-torus Gaussian regularity via confluent divided differences, Drive ID 1dn8BVsmScu690vX4Z_ewNgl0QRvMDBXcyzWiTtyJeho, proves that the exact fixed-torus limiting coordinate is a nondegenerate Gaussian Q_L with exponentially small but nonzero image corrections. The exact torus coefficient is

$$z_0, L = E[Q_L^2 \cdot 1_{\{Q_L < 0\}}] > 0,$$

not literally the planar decimal. At $L=24$ the difference is below the displayed numerical precision used in the package, but it is mathematically nonzero.

Therefore future references must use one of the following object-qualified labels:

- PLANAR_COALESCED_NORMALIZER: exact formula above;
- EXACT_TORUS_COALESCED_NORMALIZER: z_0, L , positive but not identified here by the planar decimal.

5. DEPENDENCIES AFFECTED

This closure permits the exact planar formula and decimal to be cited without reopening the derivation whenever the declared object is $Q \sim N(-b, 2)$. It supports provenance and checking of planar Palm-normalizer calculations and planar-limit diagnostic tables.

It also permanently blocks the following ambiguity:

- using 3.230978535287005... as a literal exact side-24 torus identity.

6. DEPENDENCIES UNAFFECTED

This closure does not:

- prove the exact finite- r or exact-torus normalizer coefficient numerically;
- close GP-DER-044's exact-torus regularity theorem by itself;
- prove adjacency positivity, adjacency-to-one, branch capture, or P0.2;
- change Theorem B, the live q_0 theorem state, or any machine root;
- promote a numerical value of C_{Q0} or C_* ;
- alter any historical source file.

7. TERMINAL RECORD AND REGISTER ACTIONS AFTER APPROVAL

After Dylan M. Roy approves or modifies the terminal statement:

1. Move EC-006 from Easy Closure Queue to Closure Log.

2. Record terminal label:

CLOSED — EXACT PLANAR TRUNCATED-NORMAL SECOND-MOMENT IDENTITY

3. Register object tags:

PLANAR_COALESCED**Q_NORMALIZER_SECOND_MOMENT**

$$b=6/5$$

NOT_EXACT_TORUS_DECIMAL

4. Cross-link GP-DER-044's precision correction so a cold-start reader cannot conflate the two objects.

5. Remove this identity from active re-derivation and routine review queues.

6. Preserve all cited source, audit, and correction artifacts unchanged.

8. HUMAN APPROVAL REQUIREMENT

This item is mathematically closure-ready but remains a human terminal decision under OP-PROT-004.

Proposed approval statement:

I approve GP-CLS-PROP-007-v1.0. Close the exact planar identity $z_0(b)=(b^2+2)\Phi(b/\sqrt{2})+\sqrt{2}b\phi(b/\sqrt{2})$, including $z_0(6/5)=3.230978535287004948\dots$, under the declared PLANAR_COALESCED object only. Preserve the distinct exact-torus coefficient $z_{0,L}$ and do not use the planar decimal as a literal fixed-torus identity.

9. CORRECTION PATH

If a later audit finds an algebraic, numerical, or object-definition error, do not edit this document silently. File an additive erratum identifying the exact failed line, dependency impact, and corrected terminal status.

END OF PROPOSED CLOSURE PACKAGE

Part III - Terminal-State Reconciliation

1. Closure decision

Exact internal disposition	CLOSED - EXACT PLANAR TRUNCATED-NORMAL SECOND-MOMENT IDENTITY
Closure date	2026-07-21
Evidence basis	Closed-form integration, independent quadrature, Monte Carlo corroboration, and precision/scope firewall.
Independent review already recorded	Independent formula derivation and numerical checks were performed across multiple lines.

The prior closure record is evidence about the research program's internal state, not a substitute for your review. A new material objection should be reported even if the internal label is CLOSED.

2. Correction and reopening rule

Reopen or amend this object if you identify a counterexample, an object-definition mismatch, a sign/factor/quantifier defect, an unreproducible decisive computation, a hidden premise, or a scope claim not supported by the proof. Corrections must be additive and must preserve the historical source and review record.

3. Limits

- The decimal is planar, not the exact side-24 torus coefficient; no finite-r or Palm theorem follows.
- A correct lemma may still be unusable downstream if another dependency uses a different field, law, coordinate convention, branch mark, normalization, or parameter domain.
- A prior independent review does not reduce your duty to reconstruct the argument; duplicate lineages must not be counted as additional independence.

Part IV - Reproduction Instructions

- Record your name, institution or affiliation, expertise, date, and any prior exposure to this research program.
- Reconstruct the exact statement from Part I without relying on the historical verdict.
- Audit Part II line by line. For computational objects, obtain the raw carrier from the manifest, verify hashes first, inspect source safety, then execute in a clean environment.
- Complete the obligation ledger and hostile checks. Preserve exact counterexamples, code, logs, and environment details.
- Return the verdict form below plus a finding list with severity: critical, material, clarification, editorial, or no issue.

Part V - Reviewer Verdict Form

Reviewer	_____
Affiliation / expertise	_____
Prior exposure	None / limited / substantial - explain: _____
Independent reconstruction	Complete / partial / not possible
Verdict	APPROVE / APPROVE WITH CLARIFICATIONS / AMEND REQUIRED / REJECT / CANNOT VERIFY
Exact scope accepted	_____
Critical or material findings	_____ _____
Reproduction artifacts	Code/log/hash/location: _____

Signature and date _____

Required narrative: summarize your independent derivation; list every disagreement; state whether each issue affects the theorem statement, only the proof, only reviewability/reproducibility, or only presentation. State explicitly which excluded claims you did not review.

Part VI - Manifest and Provenance

Controlling source: https://docs.google.com/document/d/1N4bU2PNUk9_3fGU1B_mAj-C7b3AT4wvT7ApoZOZ0QIU/edit

Closure / terminal record: https://docs.google.com/document/d/1Zm65j9BZe_z3P_8qP7rfZukRU-IWUE8h8bDqXdDsIQ/edit

Local source snapshot used to construct this packet: 5) normalizer formula

Local source snapshot SHA-256 (DOCX export, transport checksum only):
0d8bd541962b418b7a03ab7312a45975505a671425f03ac0ba205588c84d35c7

Authority note: the transport checksum above identifies the locally downloaded Office export used for packet construction. It is not substituted for any marker-delimited frozen-body hash declared inside the source artifact.

Supplemental Drive access may be granted for raw capsules, revision history, comments, exact native exports, and dependency artifacts. The packet is designed so that those materials strengthen reproducibility rather than supply missing mathematical context.